Class 10 Math - Chapter 6: Triangles

🌟 Introduction to Similarity

While Class 9 focused on the **Congruence** of triangles (same shape AND same size), Class 10 introduces the concept of **Similarity**.

Two figures are **similar** if they have precisely the same *shape*, but not necessarily the same *size*. For instance, all circles are similar, and all squares are similar.

Strict Conditions for Polygon Similarity:

1. Their corresponding angles are completely equal.
2. The ratio of their corresponding sides is identical (i.e., sides are proportional).

📌 Basic Proportionality Theorem (Thales Theorem)

The **BPT** is the cornerstone of this chapter.

> [!NOTE] **BPT Statement:** If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, then the other two sides are divided in the same ratio.

If DE || BC in triangle ABC, then: $$ \frac{AD}{DB} = \frac{AE}{EC} $$

The Converse of BPT

If a line divides any two sides of a triangle in the exact same ratio, then the line is directly mathematical parallel to the third side.

Mind Map of Similarity Criteria

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graph TD;
    Triangles-->Similarity_Criteria;
    Triangles-->BPT_Theorem;
    Similarity_Criteria-->AAA["AAA Similarity\n(Equal Angles)"];
    Similarity_Criteria-->SAS["SAS Similarity\n(Included Angle + Proportional Sides)
    Similarity_Criteria-->SSS["SSS Similarity\n(All Sides Proportional)"];
```

⚡ Key Similarity Criteria for Triangles

Unlike generic polygons, triangles only require *one* of the two conditions to be met to guarantee the other. The criteria for verifying $\triangle ABC \sim \triangle DEF$:

Criterion	Definition
AAA (Angle-Angle-Angle)	If corresponding angles are equal, the sides are automatically proportional.
AA (Angle-Angle)	A corollary of AAA: If two angles of a triangle equal two corresponding angles of another, the third angles must align.
SSS (Side-Side-Side)	If all three corresponding sides of two triangles are proportional, their corresponding angles are equal.
SAS (Side-Angle-Side)	If one angle of a triangle equals one corresponding angle of another, and the sides *including* these angles are proportional, they are similar.

❌ Common Mistakes

> [!WARNING]
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> 1. **Congruence vs. Similarity:** Don't swap the symbols. Congruence is $\cong$ (implies identical size), but Similarity is only $\sim$.
> 2. **Naming Convention Orders:** If $\triangle ABC \sim \triangle PQR$, it specifically means $\angle A = \angle P$, $\angle B = \angle Q$. You cannot state $\triangle ABC \sim \triangle RQP$ unless they are equilateral. The vertex order in the name matters structurally!
> 3. **Confusing BPT and Similarity ratios:** BPT compares segments of the *same* line ratio ($\frac{AD}{DB}$). Similarity compares full *sides* between two triangles ($\frac{AB}{DE}$).

📝 Quick Revision

Properties of Similar Triangles

If $\triangle ABC \sim \triangle DEF$, then:

- Ratio of perimeters = Ratio of corresponding sides
- Ratio of areas = Square of the ratio of corresponding sides $$ \frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle DEF)} = \left(\frac{AB}{DE}\right)^2 = \left(\frac{BC}{EF}\right)^2 = \left(\frac{AC}{DF}\right)^2 $$

Summary

Similarity helps us map distances over uncrossable gaps—like calculating the height of a mountain or the width of a river, just by setting up small proportional triangles. Master BPT and its converse, and always strictly pay attention to the sequential vertex naming!